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Aveo Malware Analysis

Malware Analysis

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Aveo Malware Analysis

1. Executive Summary

The submitted sample `8101c298a33d91a985a5150d0254cf426601e4632250f5a03ddac39375e7fb4d` is a 32-bit WinRAR self-extracting executable that aligns closely with publicly documented Aveo activity targeting Japanese-speaking users. Local static analysis first confirmed the file is a PE32 SFX wrapper with an appended overlay and embedded payload artefact names including `Ram.exe`, `mshelp32.exe`, `setting32.ini`, and an XLS decoy fragment beginning with `CAVE`. A follow-on controlled guest-execution workflow then directly recovered all four staged artefacts and confirmed live unpack-and-launch behaviour.

What is directly observed: the wrapper created `C:\Program Files\Ram.exe`, `C:\Program Files\mshelp32.exe`, `C:\Program Files\setting32.ini`, and `C:\Program Files\CAVE?????.xls`; `Ram.exe` executed and exposed Aveo strings including `snoozetime.info`, `hello`, `toyota`, `%APPDATA%\MMC`, `%TEMP%\MMC.EXE`, and `msnetbridge` Run-key command syntax; `mshelp32.exe` executed as a helper/cleanup component; and `setting32.ini` decoded to the lure filename `CAVE 研究会参加者.xls`. What is high-confidence assessment: `Ram.exe` is the Aveo second-stage payload, `mshelp32.exe` is a launcher/cleanup helper, and the XLS is the decoy lure. What remains reported rather than fully reproduced in-lab: successful external C2 traffic to `snoozetime[.]info` and the full interactive command cycle documented by Unit 42.

Defenders should treat this sample as a confirmed intrusion-enabler and backdoor delivery mechanism rather than a benign archive wrapper. Even though the original TRASHcan run was low-signal, the recovered second stage directly supports persistence via Run key, victim profiling, encrypted HTTP-based C2, remote shell execution, file read/write support, and broader compromise potential on any host where the extracted payload completes execution.

2. Sample Metadata

Field	Value
Family	Aveo
Malware Type	Backdoor / remote access trojan delivered via WinRAR SFX wrapper
Platform	Windows x86
Primary Sample SHA256	`8101c298a33d91a985a5150d0254cf426601e4632250f5a03ddac39375e7fb4d`
Additional Hashes	SHA1 `0146f6e1d7c7a7994cb7608f3637b8ff24f7915d`; MD5 `cd6d979280146c3205010ac3c4b81d02`
File Type / Format	PE32 GUI executable; WinRAR self-extracting archive with overlay
Delivery Theme	Japanese-language decoy spreadsheet tied to a CAVE/Ido Laboratory participation theme
First Seen / Reported	2016-08-09 public reporting; local review refreshed 2026-06-18
Analyst Confidence	Medium
Primary Objective	Establish remote access and execute operator commands

3. Source Review & Confidence

Source	Contribution	Confidence
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Internal sandbox analysis	Static triage, execution attempt, network capture, and run artefacts for this exact sample	High
Local reverse-engineering work	Confirmed PE/SFX structure, overlay-only payload artefacts, and Ghidra memory-block limitation	High
Unit 42 Aveo reporting	Corroborated family behaviour, C2, persistence, and command set for this exact SHA256	High
OTX hash page	Secondary corroboration that the sample is tracked as malicious	Medium

Directly observed evidence in this report comes from local hashes, strings, rabin2 metadata, headless Ghidra output, controlled guest-execution recovery, Sysmon-backed file/process observations, and the TRASHcan run directory. High-confidence assessment still benefits from Unit 42 reporting because the public article explicitly references this SHA256, but the critical family markers once treated as reported-only are now present directly in recovered `Ram.exe`. Remaining lower-confidence inference is limited to externally unobserved portions of the full C2 session and operator command loop that were not replayed end-to-end in-lab.

4. Initial Triage and Delivery Assessment

The primary submitted artefact is a WinRAR self-extracting executable compiled from the standard WinRAR SFX codebase (`sfxrar.pdb` path present). Local string and byte-offset review first showed the wrapper contained embedded names `Ram.exe`, `mshelp32.exe`, `setting32.ini`, and a decoy spreadsheet filename fragment ending in `.xls`. Controlled guest execution then confirmed the wrapper unpacked all four components into `C:\Program Files\` and launched both `Ram.exe` and `mshelp32.exe`. The evidence now supports a concrete split: `Ram.exe` is the Aveo backdoor stage, `mshelp32.exe` is a helper/cleanup launcher, `setting32.ini` carries the lure filename, and the XLS is the decoy document.

Variant / Artefact	Delivery Role	Notes
`8101c298...e7fb4d`	Initial SFX wrapper	Directly observed PE32 WinRAR SFX archive with appended overlay
`mshelp32.exe`	Helper / launcher / cleanup component	Recovered and executed in-lab; contains `ShellExecuteA` plus self-delete commands for `setting32.ini` and itself
`Ram.exe`	Aveo payload stage	Recovered and executed in-lab; directly contains `snoozetime.info`, `hello`, `toyota`, `%APPDATA%\MMC`, and `msnetbridge` persistence command syntax
`setting32.ini`	SFX config / lure naming component	Recovered in-lab; decodes to `CAVE 研究会参加者.xls`
`CAVE 研究会参加者.xls`	Decoy spreadsheet	Recovered as a real XLS document from wrapper drop path

5. Static Analysis

5.1. File and Build Characteristics

The file is a 241,688-byte PE32 Windows GUI executable with base address `0x400000`, four sections, and `overlay=true` according to rabin2 metadata. The debug path `d:\Projects\WinRAR\SFX\build\sfxrar32\Release\sfxrar.pdb` strongly indicates

the visible PE body is the WinRAR SFX stub rather than the Aveo implant itself. Headless Ghidra imported only the PE sections (`.Headers`, `.text`, `.rdata`, `.data`, `.rsrc`) and did not map the appended overlay, reinforcing that the interesting payload content resides after the normal image.

Static Signal	Evidence	Analyst Value
WinRAR SFX wrapper	`file` and `rabin2 -I` identify a PE32 WinRAR self-extracting archive	Explains why the visible PE body mostly reflects archive/unpack logic rather than final payload logic
Appended overlay	`overlay=true` and embedded stage filenames appear near EOF	Strong indicator that second-stage content is appended and unpacked at runtime
Embedded stage names	`Ram.exe`, `mshelp32.exe`, `setting32.ini`, `winrarsfxmappingfile.tmp`	Supports multi-file extraction and a staged lure/payload flow
Decoy lure clue	Tail bytes include `CAVE...xls`	Supports Japanese-language decoy delivery theme
SFX extraction syntax	String `-el -s2 "-d%s" "-p%s" "-sp%s"``	Confirms embedded archive extraction logic and password-capable SFX behaviour

5.2. Strings, Imports, and Embedded Artefacts

Relevant local strings in the wrapper include `Ram.exe`, `mshelp32.exe`, `setting32.ini`, `WINRAR.SFX`, `winrarsfxmappingfile.tmp`, `SetupCode`, and the extraction argument template `-el -s2 "-d%s" "-p%s" "-sp%s"`. The final bytes of the wrapper contain `setting32.ini` immediately followed by the encoded lure name that decodes to `CAVE 研究会参加者.xls`, while earlier overlay regions contain null-terminated `Ram.exe` and `mshelp32.exe` strings. Recovery of the unpacked components upgraded those observations from filename-only evidence to concrete artefacts: `Ram.exe` is a standalone PE32 compiled on 2015-08-20 with PDB path `C:\Users\SoundOF\Desktop\aveo\Release\aveo.pdb`, while `mshelp32.exe` is a smaller PE32 helper compiled on 2013-10-21. Imports in the wrapper itself remain dominated by normal WinRAR/SFX and Win32 APIs such as `ReadFile`, `GetTempPathW`, `CreateThread`, `FindResourceW`, `LoadLibraryW`, and `GetCommandLineW`.

5.3. Static Configuration Clues

The wrapper itself does not contain direct `snootetime[.info]`, `msnetbridge`, `hello`, or `toyota` strings. That separation turned out to be analytically useful rather than limiting: once `Ram.exe` was recovered, all of those markers were observed directly in the extracted payload. `Ram.exe` therefore resolves the earlier uncertainty and strongly ties the wrapper to the published Aveo persistence and C2 model instead of merely resembling it at the overlay-filename level.

6. Code Analysis

6.1. Reverse-Engineering Method

Local reversing used `rabin2`, strings, FLOSS output from the sandbox run, raw byte-offset inspection, headless Ghidra analysis on a temporary project, direct recovered-file triage, and `capa` on the unpacked executables. The headless Ghidra script confirmed the imported program entry point resolves only to CRT startup and that only normal PE sections were mapped, meaning payload-bearing overlay content sits outside the decompiler's imported memory view. The recovered files then supplied the missing behaviour-rich layer that the wrapper alone could not expose.

6.2. Functions and Logic of Interest

For the wrapper itself, only generic startup code was decompiled: `entry -> \_\_security\_init\_cookie -> \_\_tmainCRTStartup`. That result matters because it helps separate wrapper logic from payload logic and prevents overclaiming based on the visible SFX stub. The recovered executables closed that gap: `Ram.exe` imports WinINet, WinCrypt, Winsock, process-creation, and file-management APIs consistent with an HTTP backdoor, while `mshelp32.exe` is much smaller and limited to launcher/self-delete behaviour around the lure workflow.

Function / Routine	Evidence	Why It Matters
`entry`	Headless Ghidra decompiled the PE entry to CRT startup only	Confirms the imported PE body is just the SFX loader layer
SFX extraction handling	Embedded extraction syntax and mapping-temp-file strings	Supports runtime unpacking into a staging directory
Embedded file selection	Null-terminated `Ram.exe`, `mshelp32.exe`, `setting32.ini` near EOF	Strong evidence of an archive containing multiple staged components
`Ram.exe` payload logic	Direct strings/imports/capa evidence for `snoozetime.info`, RC4 markers, `msnetbridge`, `%APPDATA%\MMC`, WinINet, and WinCrypt	Confirms the second stage is the Aveo backdoor rather than a generic dropper
`mshelp32.exe` helper logic	`ShellExecuteA`, `setting32.ini`, and self-delete command strings	Supports decoy launch plus cleanup or post-extract housekeeping
Decoy naming logic	`setting32.ini` decodes to `CAVE 研究会参加者.xls`	Ties the helper flow to a real lure spreadsheet rather than a random filename fragment

6.3. Anti-Analysis, Evasion, and Obfuscation

The primary evasion feature observed locally is wrapper indirection: the operator-facing payload is hidden inside an appended SFX overlay, while the directly imported PE mostly presents as generic WinRAR code. This complicates naive static analysis and causes decompilation to focus on the stub unless the overlay is unpacked separately. The helper path also appears to include cleanup logic: `mshelp32.exe` carries self-delete commands for both itself and `setting32.ini`, reducing obvious residual artefacts after the lure and payload are launched.

7. Dynamic Analysis

7.1. Execution Overview

A deep-dive TRASHcan detonation completed under `win-malware-lab`, producing a run directory at ``/home/lost0x01/obsidian/05 Security Research/Malware Analysis/Runs/2026-06-18_131155_8101c298a33d``. That original run captured network traffic, guest artefacts, and static triage outputs, but it did not recover a malicious guest process tree, direct malicious HTTP C2, or unambiguous dropped malware files attributable to the sample. A second controlled guest-execution workflow was therefore used to watch the wrapper in real time, copy dropped artefacts before cleanup, and query Sysmon for file and process events.

7.2. Filesystem, Registry, Service, and Task Changes

The strongest dynamic evidence came from direct file creation and staging rather than the original sandbox summary. Sysmon-backed observation showed the wrapper create:

- `C:\Program Files\Ram.exe`
- `C:\Program Files\mshelp32.exe`
- `C:\Program Files\setting32.ini`
- `C:\Program Files\CAVE?????.xls`

The same execution path then showed `Ram.exe` create `C:\Users\analyst\AppData\Roaming\MMC\MMC.EXE`, which directly matches published Aveo installation behaviour. `mshelp32.exe` executed, spawned `cmd.exe`, and created `C:\Program Files\vsvarsx32.bat`, while static strings showed it also carries self-delete commands for `setting32.ini` and itself. No malicious Windows service or scheduled task was observed. The earlier `OneDrive` autorun finding from the summary remains best treated as sandbox noise rather than sample-attributable persistence.

7.3. Network and Command-and-Control Behaviour

Observed DNS and HTTP traffic in the original TRASHcan run was dominated by Windows background activity such as `www.msftconnecttest.com`, `ctldl.windowsupdate.com`, Microsoft telemetry, and SSDP discovery. No live `snoozetime[.info]` communication was captured during that window. However, this is no longer just a negative dynamic section: the recovered `Ram.exe` directly contains `snoozetime.info`, WinINet imports, and RC4-related capability markers, so the family C2 model is locally corroborated even though a full outbound session was not reproduced.

7.4. Observed Capability Summary

Capability	Supporting Dynamic Evidence
Confirmed unpack-and-drop behaviour	Wrapper created `Ram.exe`, `mshelp32.exe`, `setting32.ini`, and the decoy XLS in `C:\Program Files\`
Confirmed second-stage execution	`Ram.exe` executed and created `%APPDATA%\MMC\MMC.EXE`
Confirmed lure/helper execution	`mshelp32.exe` executed, spawned `cmd.exe`, and showed cleanup-oriented strings
No observed live C2 in-lab	PCAP and parsed network summary showed only common Windows background traffic

8. Process Tree and Execution Chain

This section is mandatory whenever execution telemetry exists.

The original sandbox summary did not recover a useful guest process tree, but the controlled guest-execution path did provide enough evidence to reconstruct a high-confidence execution chain. The chain below combines directly observed Sysmon events with recovered-file analysis.

Step	Parent Process	Child / Action	Why It Matters
1	User-launched SFX wrapper	Creates `Ram.exe`, `mshelp32.exe`, `setting32.ini`, and `CAVE?????.xls` under `C:\Program Files\`	Confirms runtime unpacking of the overlay
2	SFX wrapper / helper path	Launches `mshelp32.exe`	Supports lure orchestration and cleanup activity
3	SFX wrapper / payload path	Launches `Ram.exe`	Directly starts the Aveo second stage
4	`Ram.exe`	Creates `%APPDATA%\MMC\MMC.EXE`	Directly aligns with published Aveo install behaviour
5	`mshelp32.exe`	Spawns `cmd.exe` and creates `vsvarsx32.bat`	Indicates auxiliary launcher or housekeeping behaviour around the lure path

9. Persistence and System Modification

Local analysis now directly validates the persistence model at the string and file-staging level. `Ram.exe` contains the full command string `REG ADD HKCU\software\microsoft\windows\currentversion\run /t REG\_EXPAND\_SZ /v msnetbridge /f /d "%s"` and references both `%APPDATA%\MMC` and `%TEMP%\MMC.EXE`. Sysmon also recorded creation of `C:\Users\analyst\AppData\Roaming\MMC\MMC.EXE`, which is consistent with the reported preferred install path. A successful

Run-key write was not independently captured during this lab execution, so the registry-write portion remains strongly supported by direct payload content plus family reporting rather than by an observed E13 event.

Modification Type	Artefact	Evidence	Confidence
Run key persistence	`HKCU\software\microsoft\windows\currentversion\run\msnet bridge`	Full `REG ADD` command recovered directly from `Ram.exe`	High
Payload installation path	`%APPDATA%\MMC\MMC.exe`	`Ram.exe` strings plus Sysmon creation of `Roaming\MMC\MMC.EXE`	High
Fallback installation path	`%TEMP%\MMC\MMC.exe`	Direct `Ram.exe` string evidence	High

10. Network and Infrastructure Analysis

Stable family infrastructure from public reporting centers on `snoozetime[.info]`, with historical IP resolution to `104.202.173[.]82`, `107.180.36[.]179`, and `50.63.202[.]38`. The key update is that `snoozetime.info` is no longer only a reported family IOC in this case; it was recovered directly from `Ram.exe`. The historical IPs remain reported pivots rather than locally observed connections.

IOC / Infrastructure	Type	Role	Stability / Confidence
`snoozetime[.info]`	Domain	Aveo HTTP C2 and victim-profiling endpoint	High / High
`104.202.173[.]82`	IP	Historical resolution for family C2 domain	Low / Medium
`107.180.36[.]179`	IP	Historical resolution for family C2 domain	Low / Medium
`50.63.202[.]38`	IP	Historical resolution for family C2 domain	Low / Medium

The Aveo protocol details around HTTP GET/POST requests with RC4-encrypted parameters and a mode-change trigger on `toyota` are now partly grounded in local payload content because both `hello` and `toyota` were recovered directly from `Ram.exe`. Full wire-level reproduction of those semantics was still not achieved in this run.

11. Indicators of Compromise

The indicator picture is now split between top-level wrapper artefacts, directly recovered second-stage artefacts, and still-reported historical pivots. This materially strengthens operational confidence because the most important persistence and C2 markers are now observed in recovered payload content instead of inferred only from external reporting.

IOC Type	Example / Count	Source	Operational Notes
SHA256	`8101c298a33d91a985a5150d0254cf426601e4632250f5a03ddac39375e7fb4d`	Observed	Primary wrapper identifier
Recovered payload hashes	`Ram.exe`, `mshelp32.exe`, `setting32.ini`, `decoy.xls` SHA256 values in `iocs.csv`	Observed	Strongest validation set for extraction and follow-on hunting
Embedded and recovered	`Ram.exe`, `mshelp32.exe`,	Observed	Useful for archive-content

filenames	`setting32.ini`, `CAVE 研究会参加者.xls`		scanning and dropped-file validation
Domain	`snoozetime[.]info`	Observed	Present directly in recovered `Ram.exe`; family HTTP C2
Registry / Paths	`...\Run\msnetbridge`, `%APPDATA%\MMC\MMC.exe`, `%TEMP%\MMC\MMC.exe`	Observed	Present directly in `Ram.exe`; install path also aligned with Sysmon file creation
Historical IPs	`104.202.173[.]82`, `107.180.36[.]179`, `50.63.202[.]38`	Reported	Historical DNS pivots only

12. MITRE ATT&CK Mapping

Technique	Name	Evidence
T1204.002	User Execution: Malicious File	Delivery is a user-launched WinRAR SFX file masquerading as a document-themed lure.
T1036	Masquerading	Recovered decoy spreadsheet naming and helper flow support document-style disguise.
T1027	Obfuscated/Compressed Files and Information	The payload is embedded in a WinRAR SFX overlay rather than present as a simple standalone executable.
T1547.001	Registry Run Keys / Startup Folder	`Ram.exe` directly contains the `msnetbridge` Run-key command string.
T1071.001	Application Layer Protocol: Web Protocols	`Ram.exe` directly contains `snoozetime.info` and WinINet HTTP client imports.
T1059.003	Windows Command Shell	`Ram.exe` contains `cmd.exe`, and `mshelp32.exe` spawned `cmd.exe` during controlled execution.

13. Detection Engineering

This section contains analyst-usable detections tied to the wrapper and the now-directly-observed second-stage persistence model.

13.1. Detection Logic Summary

Signal	Telemetry Source	Analytic Goal	Tuning Notes
WinRAR SFX wrapper plus embedded and recovered stage names	File content scanning / YARA	Identify this sample and close variants	May require file-content access rather than metadata-only telemetry
`msnetbridge` Run key to `MMC.exe` in AppData/Temp	Sysmon E13 / registry telemetry	Detect Aveo persistence behaviour directly exposed in	Tune against benign software using similar names

		`Ram.exe`	
Execution of `Ram.exe`, `mshelp32.exe`, or `MMC.exe` from user-writeable paths	Process creation telemetry	Catch unpacked payload execution	Validate against legitimate internal filenames before alerting

13.2. Detection Content

```

title: Aveo Reported Persistence via msnetbridge Run Key
status: experimental
logsource:
  product: windows
  category: registry_set
detection:
  selection_key:
    TargetObject|contains: '\Software\Microsoft\Windows\CurrentVersion\Run\msnetbridge'
  selection_data:
    Details|contains:
      - '\MMC\MMC.exe'
      - '\Temp\MMC\MMC.exe'
  condition: selection_key and selection_data
level: high

```

13.3. Detection Validation and Blind Spots

The attached detections were validated for syntax and grounded in directly observed wrapper strings, recovered-payload evidence, and high-confidence public family reporting for the exact SHA256. The largest remaining blind spot is that the local lab did not capture a full successful outbound C2 session or an observed Run-key write event, so network and registry detections should still be treated as strongly corroborated hunting content rather than a fully replayed end-to-end infection trace.

14. Threat Hunting

14.1. Hunt Hypotheses

- Hosts that executed this sample or a close variant will show SFX-unpacked payload names such as `Ram.exe` or `mshelp32.exe`, and may also create `%APPDATA%\MMC\MMC.exe`.
- Hosts compromised by the Aveo second stage will show the `msnetbridge` Run value and execution of `MMC.exe` from `%APPDATA%\MMC\` or `%TEMP%\MMC\`.
- Hosts exposed to the lure path may show creation or opening of `CAVE 研究会参加者.xls` or close filename variants decoded from `setting32.ini`.

14.2. Hunt Queries

```

DeviceProcessEvents
| where Timestamp > ago(30d)
| where FileName in~ ('Ram.exe', 'mshelp32.exe', 'MMC.exe')
| where FolderPath has_any ('\AppData\','\Temp\')
| project Timestamp, DeviceName, InitiatingProcessFileName, FileName, FolderPath, ProcessCommandLine
| order by Timestamp desc

DeviceRegistryEvents
| where Timestamp > ago(30d)
| where RegistryKey has @'Software\Microsoft\Windows\CurrentVersion\Run'
| where RegistryValueName =~ 'msnetbridge' or RegistryValueData has '\MMC\MMC.exe'
| project Timestamp, DeviceName, RegistryKey, RegistryValueName, RegistryValueData,
InitiatingProcessFileName
| order by Timestamp desc

```

14.3. Triage Guidance for Hunt Hits

A meaningful hit is any Run-key creation for `msnetbridge`, any execution of `MMC.exe` from a user-writable directory outside normal Microsoft management tooling, or any host showing `Ram.exe` / `mshelp32.exe` alongside a lure spreadsheet matching the decoded `CAVE 研究会参加者.xls` pattern. Benign lookalikes may include legitimate OneDrive or software-

updater activity in AppData, but not with the same value name and path combination. Escalate to IR if the host also shows evidence of HTTP traffic to the recovered C2 domain or related file staging around decoy document execution.

15. Triage and Incident Response

- Isolate any host where `msnetbridge` persistence, `MMC.exe` from AppData/Temp, or a matching sample hash is confirmed.
- Preserve the original wrapper, any unpacked `Ram.exe`/`mshelp32.exe` payloads, registry hives, prefetch, ShimCache/AmCache, and relevant Sysmon logs.
- Scope for adjacent hosts and user activity consistent with Japanese-language lure delivery, especially archive or decoy-document execution from email or shared storage.
- Review proxy, DNS, and EDR telemetry for `snoozetime[.info` and historical IP pivots.
- Treat confirmed hits as potential broader compromise leads, not just standalone malware cleanup events, because the family includes remote shell and file-transfer capability.

16. Validation and Emulation

Validation completed in this case includes local sandbox execution, static reverse engineering of the wrapper, raw byte-offset confirmation of embedded file names, headless Ghidra inspection of the imported PE body, live recovery of the unpacked components, strings/import/capa triage of `Ram.exe` and `mshelp32.exe`, and Sysmon-backed confirmation that the wrapper dropped and launched the recovered files. No live C2 emulation or full successful outbound session replay was performed. The most concrete conclusion is now that the top-level sample is an SFX wrapper that directly unpacks a real Aveo payload, a helper/cleanup executable, a lure-naming file, and a real XLS decoy.

17. Recommendations

Priority	Owner	Action	Rationale
Critical	SOC / IR	Hunt for `msnetbridge` Run-key persistence, `MMC.exe` execution from AppData or Temp, and recovered-payload hashes from `iocs.csv`	Highest-signal observed persistence and payload clues
High	Email / Endpoint	Block or quarantine the exact wrapper hash plus recovered `Ram.exe` / `mshelp32.exe` hashes and close SFX variants carrying the same names	Reduces reinfection and catches wrapper or payload reuse
High	Threat Hunting	Search proxy and DNS logs for `snoozetime[.info` and historical pivots	Identifies potential previously compromised hosts
Medium	Detection Engineering	Deploy the attached Sigma/YARA/KQL/SPL content and tune for environment-specific noise	Improves repeat detection of wrapper, payload, and persistence behaviour

18. References and Refresh Notes

- Unit 42. "Aveo Malware Family Targets Japanese Speaking Users."
- Internal sandbox run and local reverse-engineering artefacts listed in `references.md`.
- Refresh family infrastructure before external publication because domains and historical IP mappings age quickly.